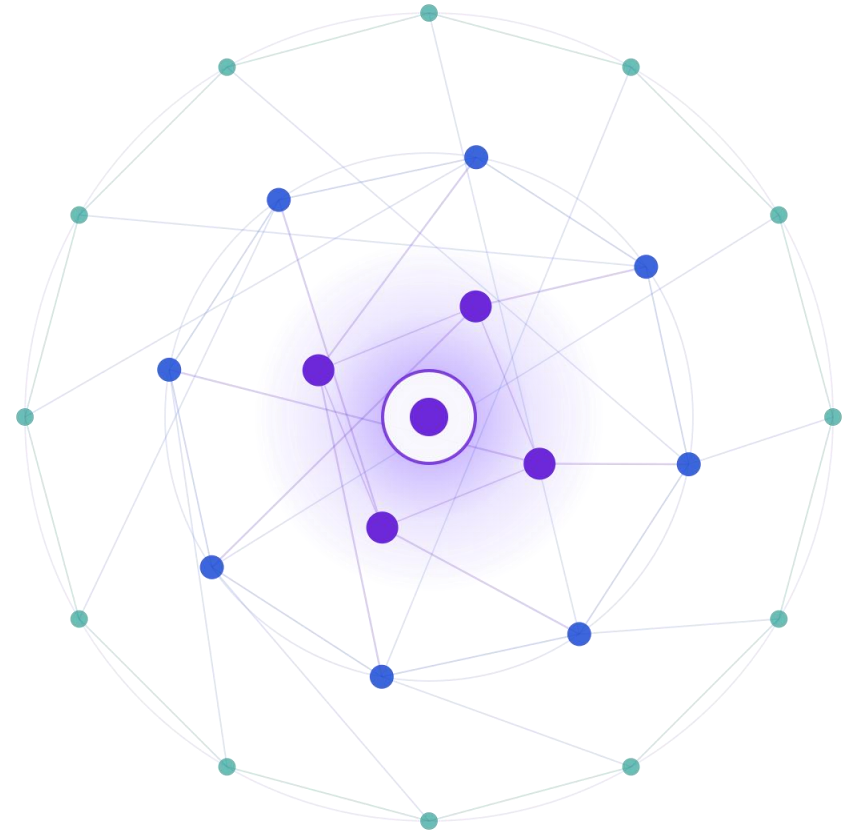


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 02 · Smart Guided Troubleshooting Engine
- Team Name - Terra
- College Name – SRM Institute Of Science and Technology
- Member Name & Email 1- Ayush Garg & aush5895@gmail.com
- Member Name & Email 2- Yash Shekhar Rai & ys9432@srmist.edu.in
- Member Name & Email 3- Yuvraj & yr9065@srmist.edu.in
- Member Name & Email 4- Daksh Kumar & dk9052@srmist.edu.in
- Submission Github link - github.com/aush5895/Prism



Theme

- **Theme 02 — Smart Guided Troubleshooting Engine**
- Customers describe faults in their own words. An agent translates that into troubleshooting steps by hand — roughly 15 minutes per scenario, across millions of calls — and the customer still has to hunt through Settings.
- What the specification asks for
 - Normalise a raw complaint into a technical query
 - Return clean, ordered troubleshooting steps as structured JSON
 - Map each step to the exact in-app Settings deeplink
 - Serve pre-validated answers from a fast-path cache in under 300 ms
- **Constraints:** Deliverable is a REST API returning JSON against a fixed Pydantic schema. Mapping must stay reusable — the production target is 10k+ scenarios.

Existing Solutions & Gaps

- Manual triage — today's process
 - An agent reads an unstructured SIIIS article, selects the relevant steps and orders them by judgement. Slow, inconsistent between agents, and the output is prose.
- A general-purpose chatbot
 - Fluent and ungrounded. It invents plausible steps, invents URLs, and cannot be audited against a source. Nothing verifies the advice came from Samsung's own text.
- Plain RAG over the knowledge base
 - Retrieves the right article, then stops. It hands the customer a list, and has no mechanism to turn a sentence into the one correct Settings screen out of 578.
- **The gap** — nobody verifies the deeplink is the RIGHT one, and nobody guides the customer through the list. We measured plain retrieval on this catalog: 31.8% of links are wrong.

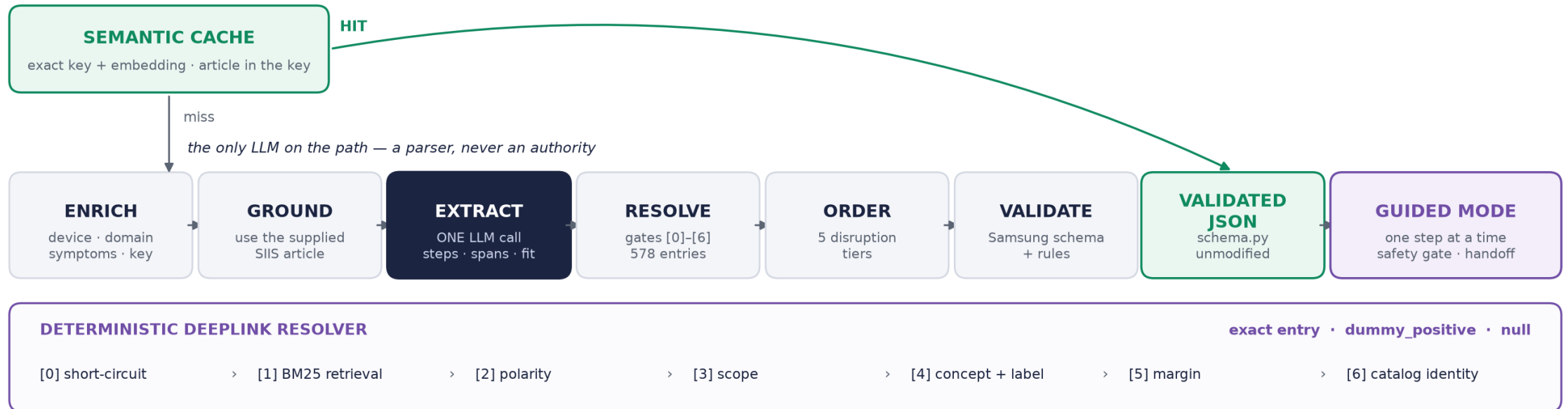
Our Solutions & Architecture Diagram

- One LLM call. Everything after it is deterministic, then guided.

POST /v1/troubleshoot

{ query, siis_response } — the Samsung article arrives WITH the request

fast path · no LLM call · searched within the supplied article only



The model never sees the deeplink catalog — the structure it returns has no field a URI could go in.

Demo & Product Walkthrough

- **1. Vague complaint in, grounded plan out**
 - “My Galaxy S22 screen inputs are delayed...” Every step is highlighted in Samsung's article; a step we cannot locate gets no span and scores nothing.
- **2. Told when the article does not fit the complaint**
 - Each problem the customer named is marked Covered, Not covered or Unverified, backed by a whole sentence quoted from the article. The gaps go to the agent.
- **3. The right Settings screen, and refusing a wrong label**
 - “Turn on” and “Turn off Touch sensitivity” resolve to different entries. A catalog entry labelled Enable Grayscale is described as mono audio, so it is refused.
- **4. Same issue, different words**
 - A reworded complaint about the same article returns from cache with no LLM call; the grounding from the cold run comes with it.
- **5. Guided, one step at a time**
 - “Did this fix it?” Before a factory reset the server stops and quotes Samsung's own warning: back up your personal data first.
- **6. Handed to an agent with the full record**
 - What was tried, each outcome, what is left. A declined step is never reported as tried, so the customer never repeats one.

Tools and tech stack used

- **Backend** Python 3.11+ · FastAPI · Pydantic v2 · POST /v1/troubleshoot, GET /health, non-spec /v1/guided/*
- **Model** Google Gemini 3.1 Flash-Lite · temperature 0, JSON-schema structured output · provider-agnostic interface with an offline deterministic fallback
- **Retrieval** rank-bm25 over message + description + qna_description · 578-entry catalog · exhaustive NumPy cosine, no ANN index needed at this scale
- **Cache** sentence-transformers all-MiniLM-L6-v2, TF-IDF fallback · exact + semantic tiers, partitioned by article · measured to 10,000 synthetic scenarios
- **Frontend** React + Vite, no component library · customer view, guided mode, engineer view
- **Quality** pytest — 265 tests, none needing an API key · seeded fuzz tests · guided flow walked in Chromium · independent review before submission
- **Deliberately not used:** FAISS, Redis, a cross-encoder reranker, LangChain, multi-agent orchestration. Each was evaluated and rejected with a stated reason — 578 entries do not need an ANN index, and a single process does not need a network cache.

Impact & Use case

- 15 minutes of manual triage becomes one API call, and a guided fix
- **Trust** schema-valid, ordered by disruption, every step traced to the article it came from, and a plain warning when that article does not match the complaint.
- **Safety** last-resort steps sit behind a confirmation, and the warning shown is Samsung's own sentence, never ours.
- **Agent time** when self-service fails, the agent receives what was already tried, so the customer is not asked to repeat anything.
- Where it is used
 - Support agent console — the plan is drafted before the agent finishes reading
 - Customer self-service — a help page that walks the customer through one step at a time
 - Samsung's catalog team — docs/catalog_health.md lists what no resolver can fix

Innovation highlights, results and limitations

- Wrong deeplinks: 31.8% → 1.2%
- 850 labelled cases. Plain BM25 retrieval scores 68.2% precision; the deterministic gates lift it to 98.7% at 88.1% accuracy@1, with 0 answers on a catalog entry whose own description contradicts its label (18 without that check). Generated by the harness, never typed.
- **Across all 20 SIIS rows:** 100% schema-valid · 100% rule-compliant · 0 URL leaks · 0 deeplinks outside the catalog · 20/20 rows answered
- **Cache:** cross-article hits impossible by construction · a 10,000-scenario run found three scaling bugs a 20-row test could not; all fixed and regression-tested
- **Limitations, stated**
 - Article fit is a model judgment with a verified quote, not proof, and is measured on only 4 mismatched pairings in 20 rows; the plan is still built from the article
 - 10 wrong answers remain, mostly a short catalog name matching inside a longer step
 - Guided mode has no resolution rate: that needs real customers
 - The gold set is derived from catalog labels, not human-labelled

What's next

- Measure resolution with real sessions
 - Guided mode already records every outcome. With real customers that becomes a resolution rate, and only then a case for reordering steps within a tier.
- Act on a mismatched article
 - Article fit already reports it. Withholding the plan on a "none" fit needs a labelled set of pairings to validate; 20 rows cannot.
- Close the subset-coverage gap
 - “Zen Mode” resolving to Screen mode. A step-side coverage term fixed 1 of the 10 remaining errors, so a stronger fix is needed.
- Multi-intent complaints
 - Samsung's own input set has three faults in one message. Splitting them is the next real capability.
- Fix the catalog with Samsung
 - 50 labels that name another screen, 5 indistinguishable groups, and no entry for Factory data reset.

Brownie points slide(differentiation)

- It says when Samsung's article does not fit
 - Everyone else trusts the supplied article. We check each problem the customer named against it, and believe "covered" only with a whole sentence quoted from it.
- Guided, not just listed
 - A server-enforced safety gate before every last-resort step, quoting Samsung's own warning. A restart is gated but never shown the factory-reset warning beside it.
- We audited Samsung's catalog
 - 50 entries name a different screen from their own description. Refusing them dropped gold-set precision from 99.2% to 98.7%. We shipped the lower number.
- We published a number that got worse, twice
 - 0 of 29 model-reported source offsets pointed at relevant text. Relocating spans by content dropped our confidence score from 0.82 to 0.76.
- Every claim is traceable
 - Design decisions cited to Samsung's spec (C1–C12), metrics generated not typed, an independent review before submission, and schema.py checksummed in CI.

Checklist- Updated on Public GitHub

- Working prototype code — public GitHub repo · Yes
- README with reproducible setup instructions · Yes, verified on a fresh clone
- Demo video, max 5 minutes · [Demo video Link](#)
- Presentation file (PPT) · this file

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India